

Joshua Liu

SKILLS

- **Programming Languages;** Python - Pygame, Java - Swing, C: Regularly used in personal projects, testing out algorithms, and university.
- **Project Management:** Managed a team of 70 plus students to successfully build a rover for the Canadian International Rover Challenge. Planned and tracked overall team progress, recruited new members through an interview process, including maintaining communication between each Division.
- **Leadership and Teamwork capabilities:** Have won multiple medals on teams I have led and been a part of.
- **Video Game Engines:** Unreal Engine 4 and 5 - Blueprints, Unity 3D - C#

EXPERIENCE

Project Manager

Schulich Space Rover Team, Calgary, AB - January, 2023 - August, 2024 & August 2025 - August 2026

- Have improved my skills in leadership, teamwork, schedule planning and conflict management.

Science Division Lead

Schulich Space Rover Team, Calgary, AB - September 2024 - August 2025

- Actively teach and learn how to write scientific reports (formal writing) and use scientific research procedures for science experiments to use at our competition.

Software Developer

Schulich Space Rover Team, Calgary, AB - September, 2022- August, 2023

- Programmed the rover control scripts, and created scripts to parse GNSS, and IMU data which was used throughout the Canadian International Rover Challenge 2023 allowing us to compete against the top teams.

Team Captain

FTC Robotics Competition - Team 16494, Red Deer, AB - September, 2019 - March, 2022

- Won a medal in 2022.
- Got 2nd in the Collins Aerospace Innovate Award in 2021.

Baseball Umpire

RDMBA - April 2015 - August 2017

- Umpired games behind home plate calling balls and strikes, and umpired games on the base path.

EDUCATION

Master's Degree, Management

University of Calgary, AB - August 2026 - Present

Bachelor of Science, Computer Science

University of Calgary, AB - September 2022 - April 2026

- Minor in Psychology
- Dean's List 2025-2026
- Signal Protocol based project for Introduction to Cryptography:
 - https://github.com/akeuben/ok_messenger
 - Programmed SHA-256, HMAC-256, HKDF, and key derivation functions with no external libraries
- An Unreal Engine 3D first person murder mystery game for my Introduction to Game Development course:
 - <https://github.com/joshua995/CPSC399Group>

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Projects

January, 2019 - Present

- **Coding Projects:** I spend time creating projects in Python and Java to enhance my knowledge of the language and understand data structures, procedural generation, simulations, and algorithms.
- **Game Development:** Have made video games using Unreal Engine and Unity - C# to work on my coding skills and problem solving techniques, while testing out new in-game algorithms. My published games are here:
<https://jl995.itch.io>

CERTIFICATIONS

DEL F B2: Proficiency in the French language, Certificate ID: 010403-202205J-6216622